Mnemos-DNA: Gated Delta-Rule Linear Attention with k-mer Fast-Weight Memory for Edge-Scale DNA Modeling

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Abstract

Gated delta-rule linear attention (Kimi Delta Attention, KDA) has been validated on natural language but, to our knowledge, never on genomic sequence modeling, where the dominant long-context architectures are implicit convolutions (HyenaDNA) and selective state-space models (Caduceus). We study it at a deliberately chosen operating point: **extremely low-parameter, edge-scale DNA language models** — ∼2.3M parameters, one 8 GB consumer GPU, megabase context — a regime that matters on its own terms for deployment on commodity and field hardware, and that makes single-axis architectural ablation affordable enough to run to completion. We pretrain a family of character-level KDA models on the human reference genome (GRCh38) at a 131,072 bp training context and ablate two additions: a per-k-mer *fast-weight memory* that attacks the finite-state recall bound of delta-rule recurrences (**Mnemos-DNA**; arm ID D06), and a coupled prototype adding one windowed local-attention layer together with a sequence-length curriculum (**Mnemos-DNA-C**; arm ID D07). Findings. (i) **At this state capacity the recall bound is not a distant cliff but a wall at distance zero**: on a repeat-copy probe over real hg38 background, a fully matched pure-KDA control (a 32×32 per-head state) shows no measurable exact recall at *any* distance (≤ 0.02 nats at 64 bp, zero beyond), and a windowed-local-attention hybrid without the memory is indistinguishable from it; the k-mer fast-weight memory converts this to 0.53–0.87 nats of copy advantage sustained to 64 kbp — degrading by only ≈0.04 nats per point substitution planted in the repeat, so the mechanism survives SNP-level divergence — for ∼3% more parameters, a ≈2× prefill-throughput cost, and a 4096× larger recurrent inference state, both of which we account for explicitly. (ii) **The address space is used, but unevenly**: in a 131 kbp window 63.1% of the 4^8 slots are ever written against 86.5% under a uniform-sequence null, with 68.5% of writes colliding and one slot (poly-T) taking 234; we give the delta rule's exact behaviour under such repeated writes. (iii) **Ultra-long fixed pretraining damages sub-kilobase grammar, and neither of our patches repairs it**: under one matched fine-tuning protocol, pretrained Mnemos-DNA reaches 0.740 on splice sites where a randomly initialized copy of itself reaches 0.832, and Mnemos-DNA-C's window-plus-curriculum leaves the gap intact (0.733); the same control applied to the released 1M-context HyenaDNA checkpoint shows the same signature (pretrained 0.737 vs. random-init 0.781 ± 0.009 on splice donors), so the effect is not specific to delta-rule architectures. (iv) **Fine-tuned architecture comparisons are protocol-fragile**: we show that letting the pooling head, epoch budget and training-set caps vary between arms — drift that arises naturally in any long-running study — produces spurious cross-arm effects of up to 30 accuracy points, an order of magnitude larger than any true architecture effect we measure; under one frozen protocol with 2–3 fine-tuning seeds, the arms differ by ≤1 point on

Code and evaluation harness: https://github.com/B4K2/Mnemos-DNA. Pretrained checkpoints (all four arms): https://huggingface.co/B4K2xx/Mnemos-DNA.

every suite mean. (v) **Perplexity is silent on all of it**: the arms sit within 0.01 nats/base of one another while the recall probe separates them by 0.6 nats and the random-init control exposes a 9-point fine-tuning gap. Every comparison to released baseline checkpoints (HyenaDNA, Caduceus — $\sim$3$\times$ our parameter count but orders of magnitude more pretraining compute) is labelled as a deployment-feasibility comparison of artifacts, not architectures.

1 Introduction

DNA foundation models face a context problem that natural-language models do not: regulatory interactions span hundreds of kilobases, yet single-nucleotide resolution matters, so a sequence of interest is routinely 10^5–10^6 tokens long. The two families that dominate this regime — HyenaDNA's implicit long convolutions [1] and the Mamba-based Caduceus [2] — replace quadratic attention with sub-quadratic mixers. A third family, *gated delta-rule linear attention* (Gated DeltaNet [3], Kimi Delta Attention [4]), has strong natural-language results but no published genomic evaluation. Its defining weakness is also its most interesting property for genomics: the recurrent state is a fixed $d \times d$ matrix, so exact recall of a distant motif is information-theoretically bounded, a recall–capacity trade-off formalized in prior work [5]. Genomes, full of near-exact repeats whose *differences* carry the signal, should be a stress test for exactly this bound.

Edge-scale is the design objective, not a constraint we apologize for. This work targets an operating point of $\sim$2.3M parameters and a single 8 GB consumer GPU (RTX 5060 Laptop). That choice is deliberate and has two independent justifications. First, it is a deployment regime with its own demand: sequencing-adjacent analysis increasingly runs on laptops, clinic workstations, and field devices, where a model that processes a megabase window in 5.6 GB is usable and a released checkpoint that goes out of memory at 512 kbp is not. Second, it is the regime in which a full single-axis ablation is actually finishable: every arm here is pretrained from scratch on a matched tokenizer, matched data order and matched token budget, which is what licenses attributing a difference to the one ingredient that changed. Large-scale reports rarely can, because they bundle architecture, budget, and data changes. We accordingly do not claim that findings here transfer to 100M+ scale, and we mark the places where the d_{model}=128 ceiling, rather than an architecture, is what a number measures.

Questions. (i) Is KDA a viable base operator for a character-level DNA model at 131 kbp training context? (ii) Where does its finite-state recall bound show up on genomic tasks, and can it be repaired within an $O(N)$ time budget — and at what cost in state? (iii) What does an ultra-long fixed pretraining context do to sub-kilobase downstream capability?

Contributions.

- **Mnemos-DNA, a k-mer fast-weight memory** (§3.1): a content-addressed memory in every layer, keyed by the exact last 8-mer, each address holding a 16×16 delta-rule fast-weight matrix [6]. It removes most of the recall wall while staying $O(N)$ in time and bitwise-exact under chunked low-memory inference.
- **An explicit collision, capacity and memory-footprint analysis** (§3.2–§3.3): what the delta rule does under repeated writes to one slot, a pigeonhole bound on collisions in a 131 kbp window, the measured departure from that bound on real hg38, and a byte-level table showing that a $\sim$3% parameter increase buys a 4096$\times$ increase in recurrent inference state.

- **An applications evaluation, not only an architecture one** (§4): pretraining, single-nucleotide-resolution classification (Genomic Benchmarks, Nucleotide Transformer), species identification across input lengths, long-range regulatory and variant-effect prediction, and the 31-task GUE suite — each with its own reading, including where our arms lose.
- **A protocol-sensitivity result for fine-tuned genomic benchmarks** (§4.2, §4.5). We quantify, on the same checkpoints, how much cross-arm variation in fine-tuning protocol (pooling head, epoch budget, training-set caps) distorts architecture comparisons: up to 30 points per task and 5 points per suite mean — spurious effects an order of magnitude larger than the real cross-arm differences, which are $\leq$1 point per suite under a single frozen protocol with 2–3 seeds. Against that frozen protocol, a matched random-init control beats *both* pretrained arms by $\sim$9 points on splice sites: fixed ultra-long-context pretraining measurably damages sub-kilobase positional grammar, and the window-plus-curriculum patch designed to repair it does not.
- **A same-harness baseline comparison, framed as deployment feasibility** (§4.9): released HyenaDNA and Caduceus checkpoints run through our identical harness at inference batch 1. They are $\sim$3$\times$ our parameter count (6.5–7.7M) with orders-of-magnitude larger pretraining budgets, and are run out-of-the-box without a chunked path, so this measures what an 8 GB deployment of released artifacts gets — not an intrinsic algorithmic advantage over Hyena or Mamba.

2 Preliminaries and Related Work

DNA sequence models. Character-level genomic language models trade the compression of k-mer or BPE tokenization for single-nucleotide resolution, which SNP- and splice-site-level tasks require. HyenaDNA [1] pretrains implicit-convolution models up to 1 M context on hg38; Nucleotide Transformer [7] and DNABERT-2 [8] take the tokenized-transformer route at much larger scale; Caduceus [2] is a bidirectional, reverse-complement-equivariant Mamba masked LM at 131 K context. We adopt HyenaDNA's tokenizer and held-out-chromosome split so that our evaluation rows are readable against theirs, and compare against *released* checkpoints through our harness, never re-implementations.

Long-context strategies. Implicit long convolutions [9], selective state-space models [10] and linear attention [11] all reduce the softmax transformer's [12] quadratic term, but they differ in what they keep: a convolution keeps a filter over relative offsets, an SSM keeps a decayed state, linear attention keeps an outer-product memory. What none of them keeps is an unbounded exact record of what was seen.

Kimi Delta Attention. KDA [4] is a linear-attention recurrence with a channel-wise forget gate, per-head write strength β, ℓ_2-normalized queries and keys, and a short causal convolution on $q/k/v$. Each head maintains a $d_v \times d_k$ state S_t (so that the retrieval $S_{t-1}k_t$ lands in value space, the correction $v_t - S_{t-1}k_t \in \mathbb{R}^{d_v}$ is well-typed, and the outer-product write $(v_t - S_{t-1}k_t)k_t^\top$ is $d_v \times d_k$) updated by a gated delta rule — an outer-product write that first subtracts what the key currently retrieves, then writes the new value — giving it error-correcting fast-weight semantics [6, 5] at $O(N)$ time and $O(1)$ state per head:

$$S_t \;=\; \mathrm{diag}(\alpha_t)\, S_{t-1} \;+\; \beta_t \left(v_t - S_{t-1}k_t\right)k_t^\top, \qquad \text{read } S_t q_t. \tag{1}$$

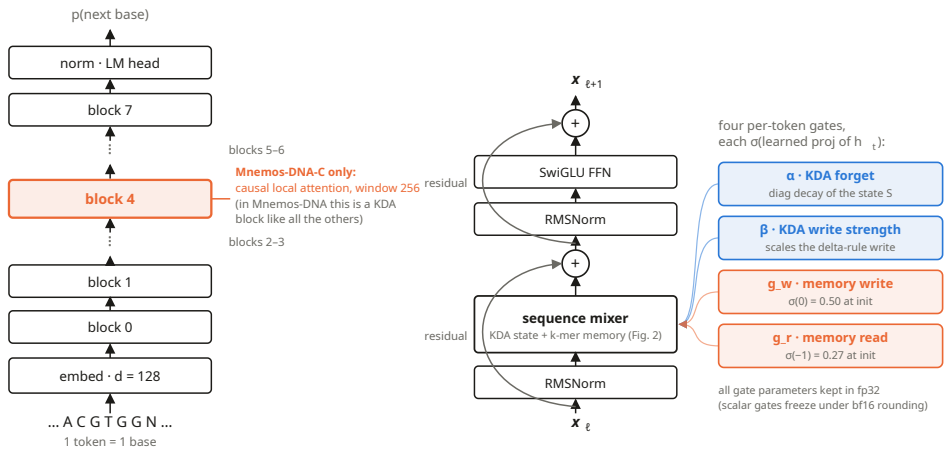


Figure 1: The Mnemos-DNA(-C) architecture, drawn vertically, input at the bottom. **(a)** Eight pre-norm blocks, d_{model}=128, 4 heads, no positional encoding (blocks 2–3 and 5–6 elided; they are identical to blocks 1 and 7). In Mnemos-DNA-C only, block 4's mixer becomes blocked causal local attention (window 256, effective span 256–511 bp). **(b)** One block's datapath: pre-norm residual structure (RMSNorm → mixer → residual, RMSNorm → SwiGLU → residual), and the four per-token gates that steer the mixer, each a sigmoid of a learned projection of the token's hidden state — the KDA recurrence's channel-wise forget gate α_t and write strength β_t, and the k-mer memory's write gate g_w (bias init 0, σ=0.50) and read gate g_r (bias init -1, $\sigma\approx$0.27). All gate parameters are kept in fp32; scalar gates freeze under bf16 rounding (§3.1). The mixer's internals are Figure 2.

The recall bound. Because S_t has fixed size, the number of (key, value) associations recoverable exactly is bounded; past that, writes interfere. In language this surfaces on synthetic multi-query associative recall [5]; here we operationalize it as a repeat-copy probe on real hg38 background (§4.6).

3 Method

3.1 Mnemos-DNA (D06): KDA with per-k-mer fast-weight memory

The backbone is 8 layers, d_{model}=128, 4 heads, vocabulary of 8 (A/C/G/T/N plus specials), no positional encoding (NoPE), 2.32M parameters; Figure 1 shows the stack and one block's datapath with its four gates, and Figure 2 the mixer's two memories. Alongside each layer's KDA mixer sits a content-addressed memory keyed by the exact identity of the last k=8 nucleotides (4^8=65,536 addresses). Each address stores not a vector but a small delta-rule fast-weight matrix (d_k=d_v=16): when position t's 8-mer address is a, the layer writes (k_t, v_t) projections of the hidden state into slot a's matrix under a learned write gate g_w (bias init 0.0, σ=0.5),

$$W_a \;\leftarrow\; W_a + g_w\big(v - W_a k\big)k^\top, \tag{2}$$

and reads slot a's matrix with the current query under a learned read gate (bias init -1.0, $\sigma \approx 0.27$), adding $g_r \cdot W_a q$ to the mixer output before the output projection. Two properties matter:

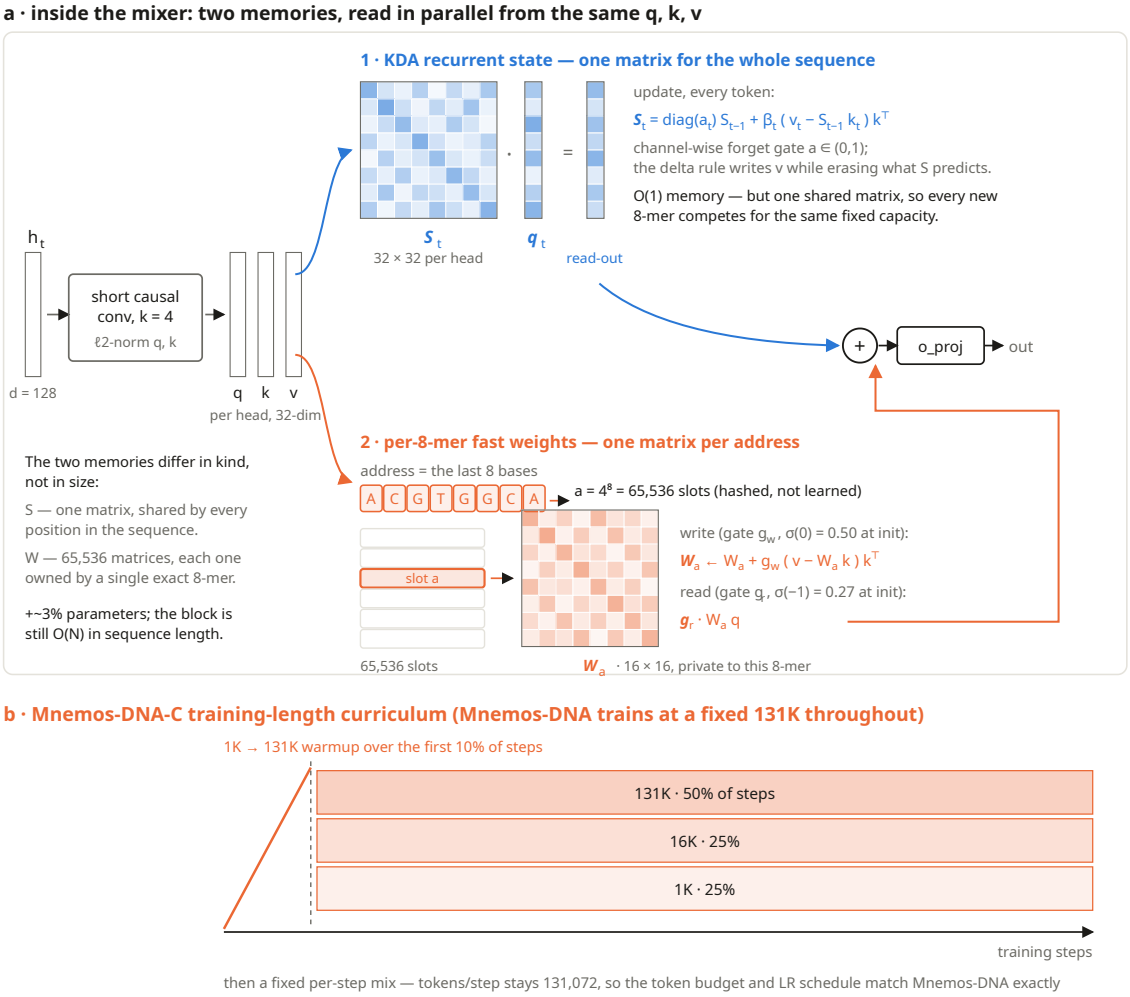


Figure 2: Inside the mixer: the two memories, and the training-length curriculum. **(a)** Two memories read in parallel from the same conv-smoothed q, k, v. *Top:* the KDA recurrence keeps one 32×32 state matrix S_t per head for the entire sequence and reads it as the matrix–vector product $S_t q_t$; the gated delta rule writes v_t while erasing what S_{t-1} already predicts, so every new k-mer competes for one fixed capacity. *Bottom:* the fast-weight memory instead holds 4^8=65,536 separate 16×16 matrices, one per exact 8-mer, and writes (g_w-gated) / reads (g_r-scaled) only the slot the current 8-mer addresses. The two reads are summed before the output projection. The table is reset per sequence and carried exactly across chunks at inference. **(b)** Mnemos-DNA-C's training-length curriculum; tokens per step are unchanged, so Mnemos-DNA and Mnemos-DNA-C share a token budget and learning-rate schedule.

- **It targets the recall bound directly.** Recall of "what followed this exact 8-mer last time" no longer competes for the KDA state's fixed capacity: the address space factors it out, and the delta rule inside each slot handles collisions of repeated 8-mers (§3.2).
- **It stays $O(N)$ in time and chunk-exact.** The full table is never materialized during the training forward — only chained per-chunk carries are computed — and chunked inference carries the table across chunks bitwise-exactly (verified at 16 K), which is what legitimizes the 1 Mbp low-memory evaluation path of §4.9. It is emphatically not free in *space*; §3.3 gives the bytes.

Two training details were load-bearing rather than incidental: gate parameters are kept in fp32 (scalar gates freeze under bf16 rounding), and the read/write gate initializations above came from a bias sweep after an earlier bug zeroed the write gate.

3.2 Collisions and capacity of the 8-mer address space

The address space is smaller than the sequences it indexes, so collisions are not an edge case — they are the normal operating regime, and the delta rule's behaviour under them is what determines whether the memory stores anything useful.

How many collisions. With $\Sigma = \{A, C, G, T\}$ there are exactly $4^8 = 65{,}536$ addresses. A window of $N = 131{,}072$ bp yields $N - k + 1$ addressed positions, so by the pigeonhole principle at least $N - 4^k = 65{,}536$ writes must land on an already-written slot, *whatever* the sequence distribution: a collision rate of at least 50%. Under a uniform i.i.d. null the expected number of distinct slots touched is $4^k\big(1 - e^{-N/4^k}\big) = 56{,}661$ (86.5% of the space), giving an expected collision rate of 56.8%. Genomic sequence is not uniform, and §4.7 measures the real figures: 41,342 slots touched (63.1%) and a 68.5% collision rate. The pigeonhole bound is therefore a floor that real DNA clears by a wide margin, and the uniform null *under*states collisions — the direction that matters.

What a collision does. Suppose slot a receives writes $(k_1, v_1), \ldots, (k_m, v_m)$ with gates $g_1, \ldots, g_m$ over one sequence, starting from $W_a^{(0)} = 0$ (the table is reset per sequence). Equation 2 gives

$$W_a^{(i)} \;=\; W_a^{(i-1)}\big(I - g_i k_i k_i^\top\big) \;+\; g_i\, v_i k_i^\top, \tag{3}$$

using $\|k_i\|_2 = 1$ (k is ℓ_2-normalized). Two regimes follow, and they are the two things the memory can be:

Proposition 1 (Exact overwrite along a repeated key)**.** *If successive writes to slot a share the same key, $k_i = k$ for all i, then along that key the slot is an exponential moving average of the written values: $W_a^{(i)}k = (1-g_i)\,W_a^{(i-1)}k + g_i v_i$. With a constant gate g, after m writes $W_a^{(m)}k = \sum_{i=1}^{m} g(1-g)^{m-i}v_i$, an EMA with effective horizon $1/g$: the slot remembers the last $\approx 1/g$ occurrences of that 8-mer and forgets earlier ones geometrically.*

Proposition 2 (Interference-free storage along orthogonal keys)**.** *If the written keys are mutually orthogonal, $k_i^\top k_j = 0$ for $i \neq j$, then $W_a^{(m)}k_j = g_j v_j$ for every $j \leq m$: all m associations coexist in the slot, up to the gate scaling. Since $W_a \in \mathbb{R}^{16\times 16}$, at most $d_k = 16$ mutually orthogonal keys fit; beyond that, writes project onto the span of existing keys and Proposition 1's overwrite behaviour applies along the overlapping directions.*

Quantity	Size	Derivation
Parameters		
Backbone (KDA, FFN, embeddings)	2.25 M	—
k-mer memory projections	66.6 K	$8 \times (4{\cdot}128{\cdot}16 + 129)$
Mnemos-DNA total	2.32 M	+2.9% over the backbone
Recurrent state carried at inference (batch 1)		
KDA recurrent state	128 KiB	8 layers × 4 heads × 32×32 × 4 B
k-mer fast-weight table (Mnemos-DNA)	512 MiB	8 layers × 65,536 slots × 16×16 × 4 B
k-mer fast-weight table (Mnemos-DNA-C)	448 MiB	7 k-mer layers; block 4 is local attention
Ratio, table to KDA state	**4096×**	

Table 1: Parameter and inference-state accounting for the k-mer fast-weight memory. Fast-weight tables and the KDA state are kept in fp32 (4 B); the recurrence is run outside autocast because scalar gates freeze under bf16 rounding. **The headline asymmetry: parameters grow 2.9% while the recurrent state a deployment must carry grows 4096×, from 128 KiB to 512 MiB.** This is the true cost of factoring exact 8-mer recall out of the fixed KDA state, and it is paid mostly in space, plus a ≈2× prefill-throughput cost relative to the matched pure-KDA control (§4.9); throughput remains flat in sequence length. A bf16 table would halve the figure to 256 MiB at the precision risk noted above. During training the full table is never materialized (only chained per-chunk carries), which is what keeps a 131 kbp training step inside 8 GB; the 512 MiB is what a streaming inference deployment holds.

Equation 3 is the recursive-least-squares/delta-rule update restricted to one address, so the general case interpolates: the slot performs gated least-squares regression of v on k over the writes it receives, retaining up to 16 well-separated key directions exactly and blending the rest as a gate-weighted EMA. This is what makes a 68.5% collision rate survivable — collisions on an 8-mer address are not collisions in the slot unless the hidden states also collide — and it also predicts where the memory should fail: when one address receives many writes with *similar* keys and *different* values, only the most recent $\approx 1/g$ survive. §4.6's decay from 0.82 to 0.62 nats between 4 kbp and 64 kbp is consistent with exactly this in-slot interference, since address collision rates are distance-independent.

3.3 Memory footprint accounting

The parameter count understates what this memory costs at inference, so we state both explicitly. Table 1 gives the byte-level breakdown.

3.4 Mnemos-DNA-C (D07): a coupled prototype (local window + length curriculum)

Mnemos-DNA-C adds two things to Mnemos-DNA at once, each motivated by a specific observed failure (§4.2): a *randomly initialized* Mnemos-DNA fine-tuned on Nucleotide Transformer splice sites beats the 4.5B-token 131 K-context checkpoint (0.832 vs. 0.740 under the final matched protocol) — pretraining at a fixed 131 K length had tuned the recurrence's timescales *away* from 300–600 bp grammar. As §4.2 reports, the patch did *not* in the end repair that failure; we describe the design here because the arm exists and its other properties (recall, efficiency) are part of the matrix.

- **Local window.** Block 4's mixer becomes blocked causal local attention with window 256 (effective causal span 256–511 bp): exact token-level recall inside the window regardless of

Arm	Mixer	Params	Position	Train ctx	Extras
D01 (Pure-KDA)	KDA only	2.25M	none	131,072	matched control
D02	KDA + local attn (1-in-4 layers, w=512)	∼2.0M	NTK-RoPE	262,144	5.0B-token budget
D06 (Mnemos-DNA)	KDA + k-mer fast-weight memory	2.32M	none	131,072	—
D07 (Mnemos-DNA-C)	Mnemos-DNA, block 4 → local attn (w=256)	2.29M	none	131,072	length curriculum

Table 2: The arms analysed in this paper. D01 (Pure-KDA) is the KDA-only control, fully matched to Mnemos-DNA(-C): NoPE, 131 K training context, 4.5B tokens, same data order. (An earlier pure-KDA arm, D04, was excluded after an audit found its bf16 scalar cache gates had never trained; Appendix B.) D02 is the window-only control — windowed local attention *without* a k-mer memory — and appears in the language-model sweep (Figure 4), the recall probe (Figure 9), the long-range probes (Table 4) and the embedding analysis (Figure 7). D02 predates the Mnemos-DNA(-C) protocol freeze: it was trained with NTK-RoPE [13, 14] at a 262 K context for a 5.0B-token budget, so its rows are read as a window-only reference point, not as a member of the matched Mnemos-DNA(-C) pair. Other earlier ablation arms of the wider study (a kernelized-linear-attention arm, an evicted-KV cache, a vector-valued k-mer memory) established the design and are referenced where relevant.

learned timescales. Window 256 rather than 512 was chosen so that a 300–600 bp task sequence is not resolvable in a single exact hop.

- **Length curriculum.** Training length warms up 1 K→131 K over the first 10% of steps, then draws per-step lengths from a fixed mix {1K: 0.25, 16K: 0.25, 131K: 0.5}. Tokens per step are unchanged (131,072), so the token budget and learning-rate schedule are identical to Mnemos-DNA. A loss-driven length controller was considered and rejected: bang-bang switching oscillates and is not reproducible across arms; the fixed mix keeps both regimes in the loss by construction, monitored by long- and short-context validation losses.

Mnemos-DNA-C is a coupled prototype, and we do not claim otherwise. Because the window and the curriculum were introduced together, any result attributed to Mnemos-DNA-C in this paper is a property of the pair. The single-axis controls exist as configurations — curriculum without window, and Mnemos-DNA itself (neither) — but the curriculum-only arm was untrained at this preprint's checkpoint. Under the matched fine-tuning protocol of §4.2 the pair turns out to change downstream means by ≤1 point, so the attribution question has become less urgent than initially assumed; what the pair *does* measurably change is long-range recall (slightly down, §4.6) and short-context validation loss (held within 0.03 nats of long-context, Figure 3).

Micro-batching is capped by tokens in flight, not sequences ($4 \times 32{,}768$ peaks at 6.17 GiB where $1 \times 131{,}072$ peaks at 4.35 GiB), which is what lets the curriculum train on the 8 GB card. Mnemos-DNA-C has 2.29M parameters.

4 Experiments

Pretraining setup. Human reference genome GRCh38/hg38 [15], single-nucleotide tokenization (no k-mers, no BPE; vocabulary 8), HyenaDNA's held-out-chromosome split, causal next-nucleotide objective. Token budget 4.5B (34,177 steps × 131,072 tokens), AdamW, peak learning rate 10^{-3} with cosine decay to 0.1×, warmup 200 steps, bf16 with gradient checkpointing. Hardware: one NVIDIA RTX 5060 Laptop GPU (8 GB, Blackwell). **The checkpoints analysed are trained on 4.5B tokens** — step 34,177 (Mnemos-DNA), 33,900 (Mnemos-DNA-C) and 34,408 (Pure-KDA, D01) — and every number below comes from those checkpoints.

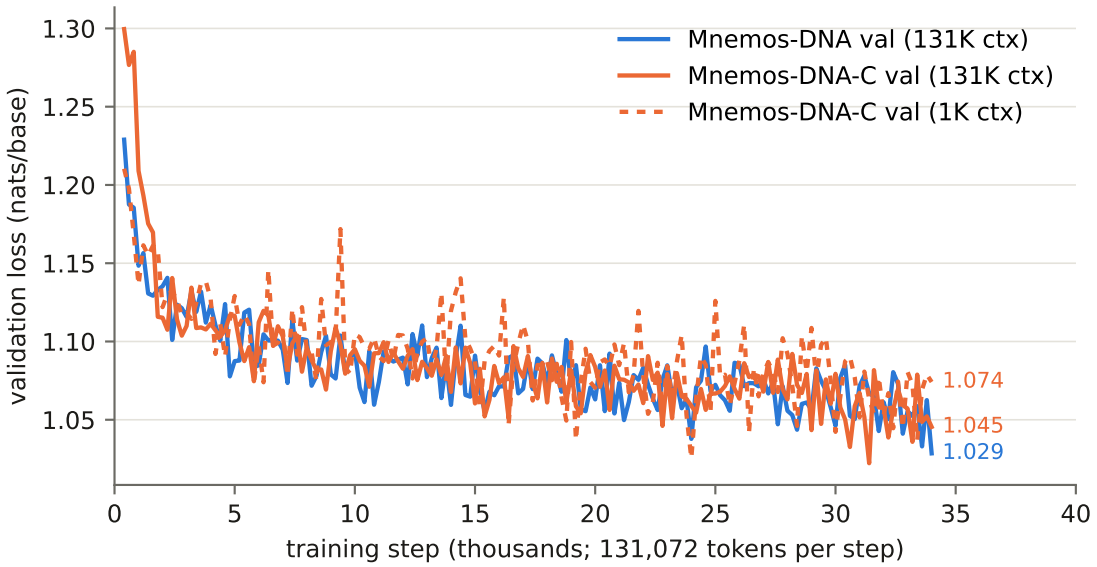


Figure 3: Validation loss. Mnemos-DNA and Mnemos-DNA-C track each other closely at the shared 131 K evaluation length (final 1.029 vs. 1.045 nats/base); Mnemos-DNA-C additionally holds a 1 K-context validation loss within 0.03 nats of its long-context loss, the curriculum's explicit target. Curves start after learning-rate warmup.

Evaluation harness. Every number comes from one suite: per-(benchmark, context) reseeding of numpy and torch, a 4M-token floor for language-model evaluation per context length, 128 trials per needle point, and a percentile bootstrap (n=1000) on every test-set metric — resampling per prediction (classification), per window (LM; per-token bootstraps understate the interval by an order of magnitude), or per trial (needle). These intervals cover split-sampling uncertainty only, not seed variance. Downstream tasks run in two modes: a frozen-backbone linear probe (a lower bound, labelled as such) and full fine-tuning (the protocol family of HyenaDNA / Nucleotide Transformer / DNABERT-2; effective batch 32, up to 6 epochs / 6,000 steps, best-dev-epoch selection, a [mean | last | 8 positional bins] head pooled over forward and reverse-complement passes). External baselines — `hyenadna-large-1m`, `hyenadna-medium-450k`, `caduceus-ph/ps` (131 K) — run through the same harness with vocabulary remapped losslessly in both directions. Caduceus is a bidirectional masked LM, so causal LM and needle metrics are skipped for it by design. The chunked low-memory path is used only for our KDA arms, where it is verified bitwise-exact; the external checkpoints are run out-of-the-box, so their out-of-memory ceilings describe those released artifacts as shipped (§4.9).

4.1 Pretraining on the human genome

Validation loss tracks closely between the arms throughout training (Figure 3). At 4.5B tokens, held-out cross-entropy at 131 K context is 1.065 (Mnemos-DNA), 1.067 (Mnemos-DNA-C) and 1.074 (Pure-KDA, D01) nats/base — 1.54 bits/bp, perplexity 2.90 over the 4-letter alphabet — and stays within ±0.03 nats from 8 K to 1 M bp of evaluation context for all arms (Figure 4). The k-mer memory buys ≈0.005–0.01 nats over the matched pure-KDA control across the sweep, and the local window and curriculum are loss-neutral on top of it: the three arms are at near-*parity*. Whatever separates them elsewhere (§4.6, §4.2), it is not next-nucleotide modeling of megabase windows. This is the first of several places where perplexity carries no signal about the differences

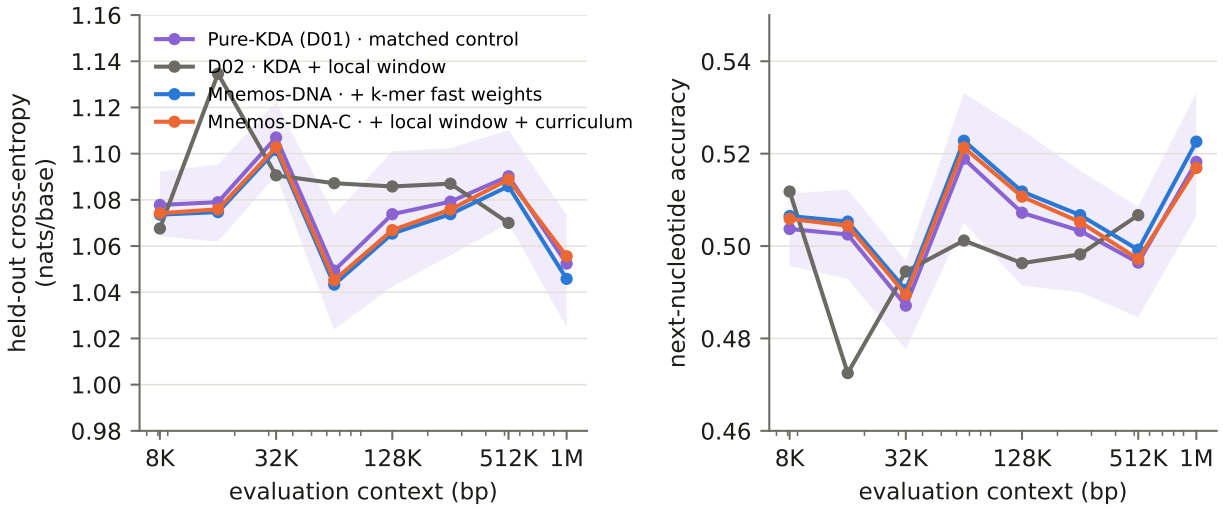


Figure 4: Held-out language-model quality does not degrade with evaluation context, from 8 kbp to 1 Mbp. Left: cross-entropy (a 4 M-token floor per point; scatter is genomic-window variation, not a trend). Right: next-nucleotide accuracy against a 0.25 chance rate. The k-mer-memory arms (Mnemos-DNA(-C)) sit ≈0.005–0.01 nats below the matched Pure-KDA control (D01) and ≈0.02 below the windowed hybrid (D02, 5.0B tokens). All arms train on one 8 GB GPU.

that matter downstream.

4.2 Single-nucleotide resolution: Genomic Benchmarks and Nucleotide Transformer

Protocol sensitivity, measured on the same checkpoints. Because fine-tuning protocols in long-running studies drift — a better pooling head or a longer epoch budget is typically adopted for the newest arm first — we can quantify what such drift does to an architecture comparison, using this study's own history as the experiment. With Mnemos-DNA fine-tuned under an older protocol ([mean|last] head, 3 epochs) and Mnemos-DNA-C under the current one (bins+RC head, 6 epochs), the identical checkpoints produce 0.618 vs. 0.922 on splice sites — an apparent +30.4-point architecture effect. Under one matched protocol the same comparison is 0.740 vs. 0.733. On GUE, capping both arms' training sets identically (vs. full-data for one arm on 28 of 31 tasks) moves the apparent suite-mean gap from −5.3 points to −0.1. Every downstream number in this paper therefore uses one frozen protocol (6 epochs / 6,000 steps, 4,000/2,000 caps, bins+RC head, 2–3 fine-tuning seeds per cell), launched from a single queue script that records the full command line per row — including the random-init control, which is run under the same head as the arms it is compared against.

What the matched rerun shows (Figure 5, Table 3). Three things:

- **The pretraining damage is real and survives the audit.** A randomly initialized Mnemos-DNA fine-tuned under the same head reaches 0.832 ± 0.002 on splice-sites-all and 0.897 ± 0.005 on splice donors, where the pretrained arms reach 0.71–0.79. On the position-sensitive splice tasks, 4.5B tokens of fixed-131 K pretraining make the model *worse* than no pretraining — by ∼9 and ∼11 points respectively. On the remaining Nucleotide Transformer

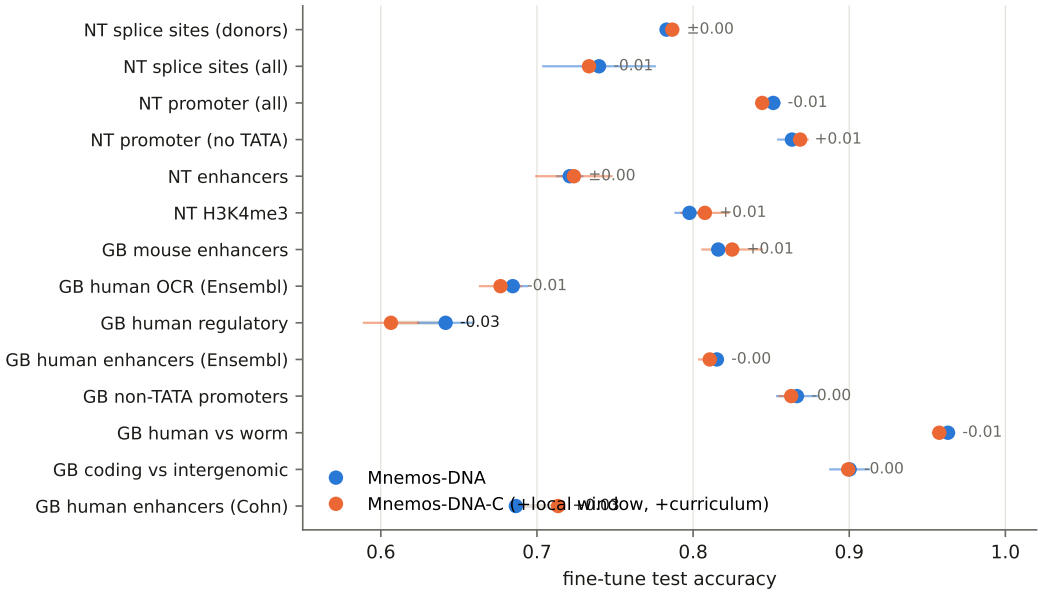


Figure 5: Full fine-tuning, Mnemos-DNA → Mnemos-DNA-C under the matched protocol, Genomic Benchmarks (8 tasks) and the 6-task Nucleotide Transformer subset; dots are seed means, whiskers ±1 sample std across fine-tuning seeds. No per-task difference exceeds 3.5 points and most sit inside the seed spread. Under an *unmatched* protocol the same checkpoints show spurious gaps of up to 30 points on this panel; see the protocol-sensitivity analysis below.

Task (full fine-tune, test accuracy)	Pure-KDA (D01)	Mnemos-DNA	Mnemos-DNA-C	random init
NT splice sites (donors)	0.779	0.783 ±0.001$^{(2)}$	**0.787 ±0.002**$^{(3)}$	0.897 ±0.005$^{(2)}$
NT splice sites (all)	0.713	**0.740 ±0.036**$^{(2)}$	0.733 ±0.003$^{(3)}$	0.832 ±0.002$^{(2)}$
NT promoter (all)	0.835	**0.851 ±0.004**$^{(2)}$	0.844 ±0.002$^{(3)}$	0.850 ±0.003$^{(2)}$
NT promoter (no TATA)	0.865	0.863 ±0.009$^{(2)}$	**0.869 ±0.005**$^{(3)}$	0.846 ±0.006$^{(2)}$
NT enhancers	**0.738**	0.721 ±0.008$^{(3)}$	0.724 ±0.024$^{(3)}$	0.735 ±0.010$^{(2)}$
NT H3K4me3	0.792	0.798 ±0.009$^{(3)}$	**0.808 ±0.015**$^{(3)}$	0.790 ±0.002$^{(2)}$
GB mouse enhancers	**0.831**	0.816 ±0.003$^{(2)}$	0.825 ±0.019$^{(3)}$	—
GB human OCR (Ensembl)	**0.694**	0.684 ±0.009$^{(2)}$	0.677 ±0.013$^{(3)}$	—
GB human regulatory	0.608	**0.641 ±0.018**$^{(2)}$	0.607 ±0.017$^{(3)}$	—
GB human enhancers (Ensembl)	0.799	**0.815 ±0.002**$^{(2)}$	0.811 ±0.007$^{(3)}$	—
GB non-TATA promoters	0.836	**0.867 ±0.013**$^{(2)}$	0.863 ±0.008$^{(3)}$	—
GB human vs worm	0.962	**0.963 ±0.004**$^{(2)}$	0.958 ±0.004$^{(3)}$	—
GB coding vs intergenomic	0.896	**0.900 ±0.012**$^{(2)}$	0.899 ±0.002$^{(3)}$	—
GB human enhancers (Cohn)	0.710	0.686 ±0.002$^{(2)}$	**0.714 ±0.002**$^{(3)}$	—

Table 3: Absolute per-task fine-tuned accuracy on the Nucleotide Transformer subset and all eight Genomic Benchmarks tasks under one matched protocol for every column (6 epochs / 6,000 steps, 4,000/2,000 train/test caps, [mean|last|8 bins]+RC head). Mnemos-DNA(-C) cells are mean ± sample std over fine-tuning seeds (per-cell n in superscript; Mnemos-DNA's seed-0 Genomic Benchmarks rows were archived by the harness's fingerprint guard, leaving n=2 there). Pure-KDA (D01) is a single seed. The random-init column fine-tunes the Mnemos-DNA architecture from scratch under the same head (2 seeds, Nucleotide Transformer only). All our arms are ∼2.3M-parameter models pretrained for 4.5B tokens on a single 8 GB consumer GPU; published HyenaDNA / Nucleotide Transformer / DNABERT-2 numbers on these suites come from larger models and far larger pretraining budgets, so any comparison to them is a comparison across compute budgets, not architectures.

tasks the pretrained arms are at or slightly above the random-init control, so the damage is specific to sub-kilobase positional grammar, not a general failure to transfer.

- **No repair under the matched protocol.** Mnemos-DNA-C's window-plus-curriculum moves splice-sites-all from 0.740 to 0.733 and splice donors from 0.783 to 0.787 — noise. Across all 14 tasks the arms differ by at most 3.5 points in either direction, and the suite means differ by $\leq$0.3 points. The +30.4 was the head and epoch budget, not the architecture.
- **The matched pure-KDA control lands in the same band.** D01, with no k-mer memory at all, is within $\sim$2 points of the memory arms on nearly every task. Whatever the fast-weight table contributes to recall (§4.6), these 200–600 bp classification tasks do not exercise it.

The damage is not specific to our architecture. To test whether the pretraining-vs-random-init inversion is a KDA quirk, we ran the same control on the released HyenaDNA-1M checkpoint: the pretrained artifact and a randomly initialized copy of the identical architecture (6.5M parameters, same tokenizer), both fine-tuned on the Nucleotide Transformer suite under the same matched protocol (random init: 2 seeds). The 1M-context pretrained checkpoint gains nothing over random init on any of the six sub-kilobase tasks, and on splice donors it falls clearly below (0.737 vs. 0.781 ± 0.009; splice-sites-all 0.655 vs. 0.689 ± 0.038, the same direction but within seed spread). The effect is milder than on our arms (4 points vs. 9–11) and rests on one pretrained seed, so we state it as: the signature generalizes — ultra-long fixed-context pretraining fails to help, and on the most position-sensitive task hurts, sub-kilobase grammar for an implicit-convolution architecture as well. Note these rows describe HyenaDNA's *1M-context* checkpoint under our matched protocol and caps; they are not comparable to HyenaDNA's published short-task numbers, which use its short-context (1 K) checkpoints — a deployment choice by its authors that is itself consistent with this finding.

Two ceilings are worth naming rather than burying. Nucleotide Transformer `promoter_all` is pinned at 0.84–0.85 for every arm *and* for the random-init control, while GC content alone yields 0.81: that is a d_{model}=128 capacity ceiling, not an architecture verdict. And the splice-site gap to random init is the sharpest open problem this study leaves: the curriculum-plus-window patch that was designed to close it does not.

4.3 Species classification across input lengths

Species identity is the task where more context monotonically helps: accuracy rises to 1.000 at 512 K–1 Mbp windows for Mnemos-DNA (Figure 6). The sweep embeds at batch 1, the deployment-relevant setting, where the externals reach 128–256 K; note that a batched embedding pipeline (8 sequences at once) instead drives them out of memory at 16–32 K, overstating the length gap by nearly an order of magnitude — the ceilings reported anywhere in this paper are single-sequence. Since all models saturate near 0.99 by 16 kbp, the accuracy axis is not where the arms separate — the length axis is, and that separation is a statement about deployable window size, discussed as such in §4.9.

What the embeddings look like. Figure 7 projects mean-pooled final-layer embeddings of 8 kbp windows with t-SNE [16], one panel per arm (D02, Mnemos-DNA, Mnemos-DNA-C). All three arms, trained only on *human* sequence with a causal next-nucleotide objective, organize five species into linearly separable clusters — which is why the frozen probe of Figure 6 works at all. The structure is consistent across arms: human, mouse and zebrafish (vertebrates) form adjacent

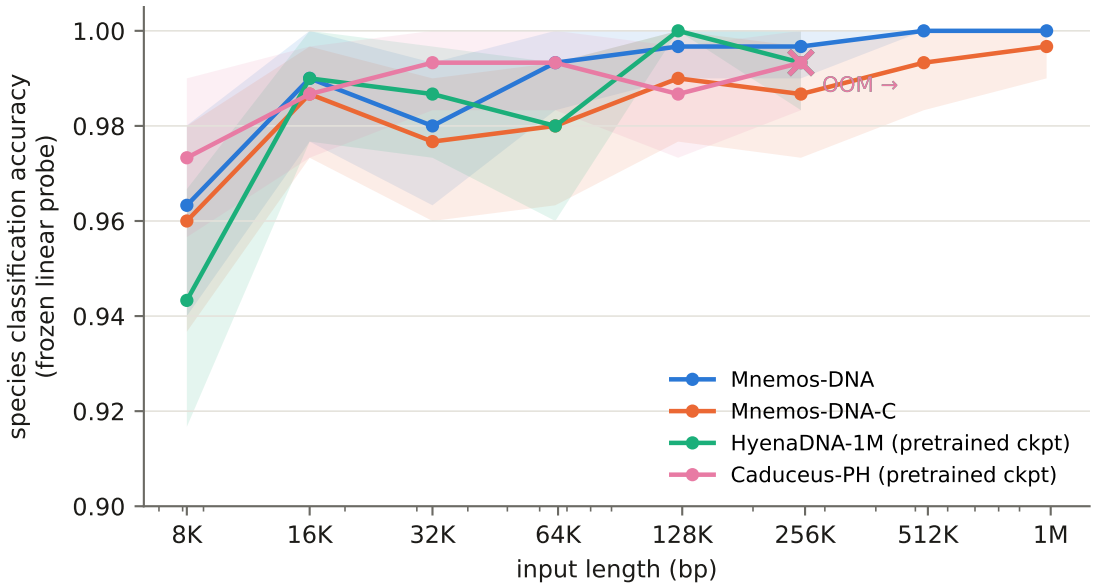


Figure 6: Species classification versus input length (frozen linear probe, batch-1 embedding, 5-way: human/mouse/zebrafish/fly/worm; shaded bands are 95% bootstrap CIs; × marks the last length before OOM). All models saturate near 0.99 by 16 K; the differentiator is how far each can go — Mnemos-DNA reaches 1 Mbp windows at 1.000 while the released external checkpoints stop at 256 K (Caduceus-PS at 128 K). **The length ceilings are a deployment-feasibility comparison of released out-of-the-box checkpoints on consumer hardware, and confound model size with architecture**; the external models are ∼3× our parameter count with far larger pretraining budgets and are run without a chunked path.

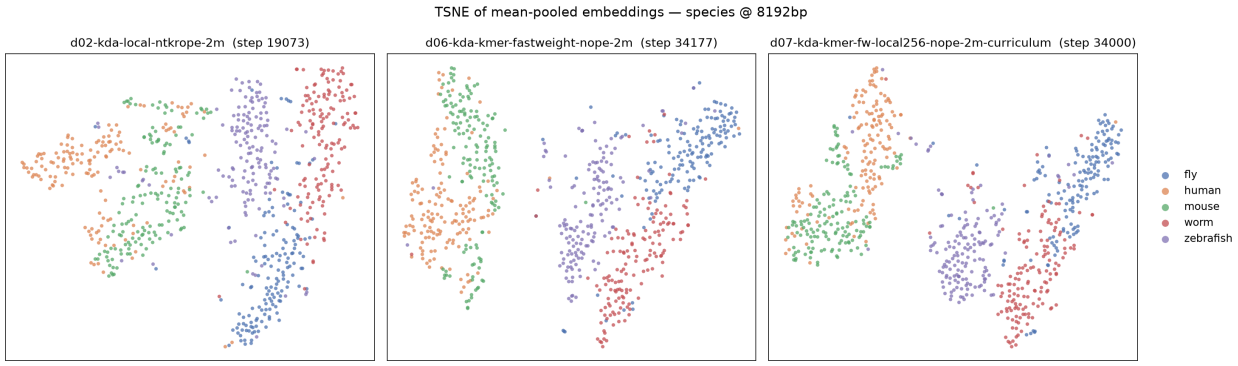


Figure 7: t-SNE of mean-pooled embeddings of 8,192 bp genome windows from five species (human/mouse/zebrafish/fly/worm), one panel per arm — D02 (window-only), Mnemos-DNA, Mnemos-DNA-C — all pretrained on human sequence only. Species separate without any species supervision, and vertebrates cluster together, mirroring phylogeny; this is the representation the frozen probe of Figure 6 reads out.

clusters while fly and worm sit apart, i.e. the embedding geometry loosely recapitulates phylogenetic distance rather than being an arbitrary partition. The k-mer-memory arms (Mnemos-DNA(-C)) show modestly tighter vertebrate separation than the window-only D02, but all three are already probe-saturated at this length, so we present the panel as a qualitative account of *why* the probe saturates, not as an arm comparison.

Task (probe, 8 K windows)	D01	Mnemos-DNA	Mnemos-DNA-C	D02	HyenaDNA-1M	Caduceus-PH	Caduceus-PS
GLRB VEP eQTL (acc)	0.664	**0.672**	0.625	0.664	0.586	0.586	**0.672**
GLRB enhancer (acc)	0.616	0.608	0.648	**0.664**	0.592	0.624	0.632
GLRB promoter (acc)	0.942	0.942	0.942	**0.950**	0.934	0.934	0.934
GLRB chromatin (mean acc)	—	0.931	0.937	0.929	0.933	0.937	**0.938**

Table 4: Genomics Long-Range Benchmark, frozen-backbone linear probes at 8 K context, same harness for every column; D01 is the matched pure-KDA control and D02 the window-only control of Table 2. A probe is a lower bound on all models equally. External checkpoints (right three columns) are ∼3× our parameter count and pretrained with orders of magnitude more compute than ours; typical 95% CIs span ±0.07 on eQTL/enhancer, so most differences here are individually indistinguishable — the honest summary is parity, reached at ∼2.3M parameters.

Suite (fine-tune, mean acc)	#tasks	Pure-KDA (D01)	Mnemos-DNA	Mnemos-DNA-C	Δ(C−M)
Genomic Benchmarks [18]	8	0.792	**0.797**	0.794	−0.3
Nucleotide Transformer subset [7]	6	0.787	0.793	**0.794**	+0.1
GUE (DNABERT-2) [8]	31	—	**0.750**	0.749	−0.1

Table 5: Suite-level mean fine-tuned accuracy under the matched protocol (Genomic Benchmarks and Nucleotide Transformer means are over the seed means of Table 3; GUE is seed 0 for both arms; D01 was not run on GUE). GUE data is the `leannmlindsey/GUE` HuggingFace community mirror of the DNABERT-2 release (the official distribution cannot be fetched non-interactively), and should be cited as a community mirror. Note the GUE means are not comparable to published DNABERT-2 numbers: the 4,000-example training cap that makes the arms comparable to *each other* also trains on far less data than the official protocol on the larger tasks.

4.4 Long-range regulatory and variant-effect prediction

Table 4 gives the frozen-probe results on the Genomics Long-Range Benchmark [17] at 8 K windows: at ∼2.3M parameters our arms sit at parity with the external checkpoints on all four task groups, with most differences inside the bootstrap intervals. The more interesting result is what happens when the window grows. Where species identity rewards context, variant effect punishes it: every model degrades from 8 K to 32 K windows on GLRB VEP eQTL (Mnemos-DNA: 0.672 → 0.578; Table 4 reports the 8 K point). Diluting a SNP-centered signal with context is not free. "More context helps" is a per-task question, and this is the task where the answer is no — worth stating in a paper whose efficiency story is about long windows.

4.5 GUE at the matched protocol, and a regression that dissolved

Under the matched protocol the GUE ledger closes at zero: mean 0.750 (Mnemos-DNA) vs. 0.749 (Mnemos-DNA-C), with every per-task difference within ±3.8 points (Figure 8, Table 5). The unmatched-protocol picture had suggested two further conclusions; neither survives the matched rerun, and dissecting why is instructive.

`virus_covid` — under the unmatched protocol an alarming 0.657→0.177 regression that a learning-rate sweep could not recover — is explained: the 0.657 came from fine-tuning on the task's full 73,335-example training set, the 0.177 from the 4,000-example cap. Under matched caps both arms sit at 0.166–0.177 on this 9-class task, barely above the 0.124 majority baseline, with near-maximal prediction entropy. The correct statement is that *neither* arm learns SARS-CoV-2 variant classification from 4,000 examples; the learning-rate sweep was probing a data-budget artifact. The same applies to the `emp_*` yeast epigenetic-mark "degradation": it tracked training-set size, not

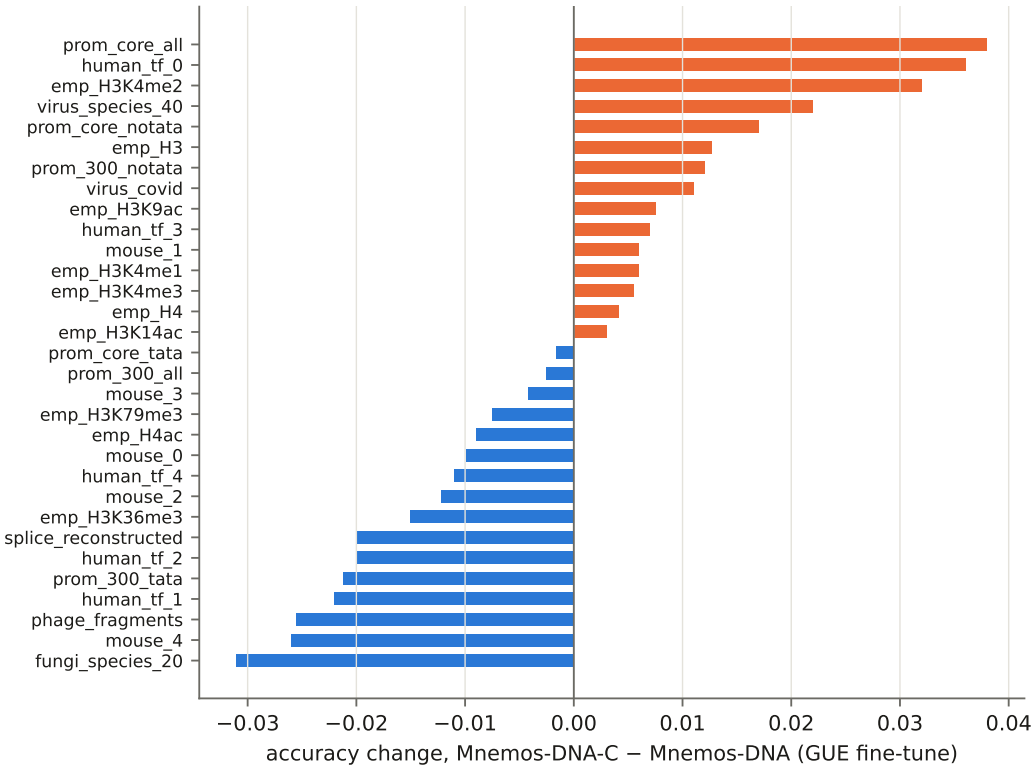


Figure 8: Per-task accuracy change on the 31-task GUE suite under the matched protocol (both arms: 4,000-example training cap, 6 epochs, bins+RC head, seed 0). The largest per-task difference in either direction is ±3.8 points and the suite means differ by 0.06 points — the arms are equivalent on GUE. Under an unmatched protocol (differently sized training sets on 28 of 31 tasks) the same checkpoints instead show a spurious −5.3-point mean loss for Mnemos-DNA-C, dominated by the yeast `emp_*` tasks.

architecture.

What survives is the dissociation: perplexity predicted none of this in either direction — the arms are 0.002 nats apart (§4.1), appear 5–30 points apart per task under an unmatched protocol, and are ≤4 points apart under the matched one. Downstream deltas of that size are within what *evaluation protocol variation alone* produces, which is itself this domain's restatement of "perplexity is not capability" — and a caution about how easily fine-tuned comparisons manufacture architecture stories.

4.6 Mechanism I: the recall bound

The mechanistic result resolves cleanly (Figure 9), and it now rests on a fully matched control: D01 shares Mnemos-DNA's positional scheme (NoPE), training context, token budget and data order, and differs only by the k-mer memory. Pure KDA's copy advantage is ≈0.02 nats at distance 64 and zero beyond — on genomic background it never performs measurable exact recall at *any* distance, so the finite-state recall bound is not a distant cliff here but a wall at distance zero. The window-only arm D02, rerun in copy mode for this figure, traces the pure-KDA curve point for point (0.016 at 64 bp, zero beyond): windowed local attention, despite exact token-level access inside its 512 bp span, does not by itself convert a repeated segment into a measurable copy advantage on hg38 background. The fast-weight memory (Mnemos-DNA) recovers 0.87 nats at short range and still

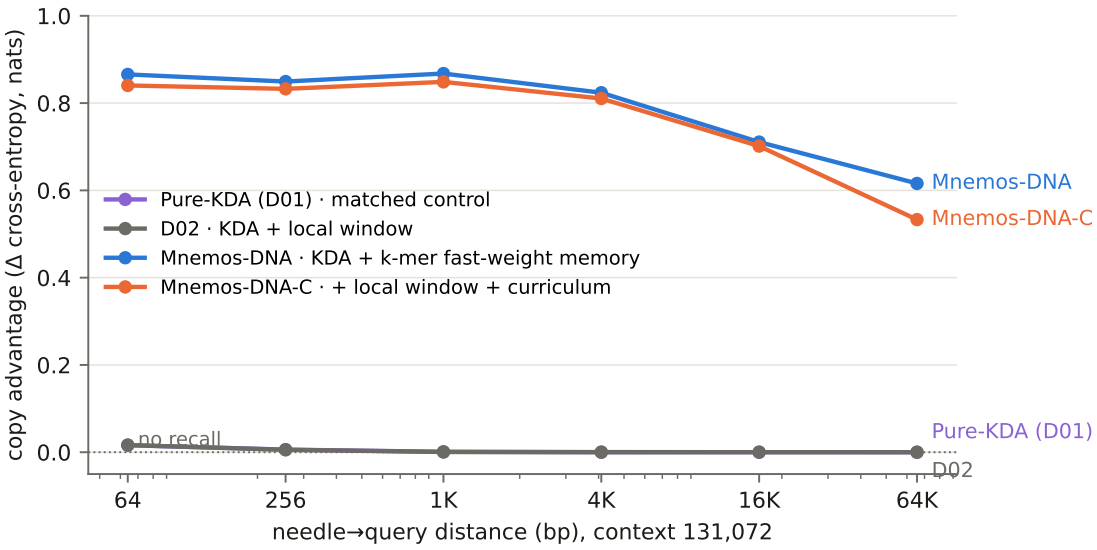


Figure 9: Long-range exact recall (context 131,072). A 256 bp segment of real hg38 sequence is planted and the model must lower its loss when the segment repeats a distance d later; the y-axis is the paired per-trial cross-entropy reduction versus a control (128 trials/point). The matched Pure-KDA control (D01) and the window-only hybrid (D02) are indistinguishable from zero beyond 64 bp. The k-mer fast-weight arms hold 0.53–0.87 nats of advantage out to 64 kbp, decaying smoothly with distance.

holds 0.62 at 64 kbp. Mnemos-DNA-C sits slightly below Mnemos-DNA at long distances (0.53 at 64 kbp) — the curriculum's short-sequence steps plausibly de-emphasize ultra-long carry — but the two memory arms are qualitatively identical. The memory, not the window, is what buys recall.

The wall is a statement about this capacity, not about delta rules in general. We emphasize that the distance-zero wall is measured at edge scale: with d_{model}=128 over 4 heads, each KDA head's state is a single 32×32 matrix — 1,024 floats asked to summarize up to 131,072 tokens of genomic background, a regime where sustained exact recall of an arbitrary planted segment is information-theoretically excluded long before any architectural subtlety matters. State capacity grows quadratically in head dimension, and larger delta-rule models on language do exhibit a graded recall-versus-distance profile rather than a wall [5, 3]. The claim this section licenses is therefore bounded: *in the low-parameter edge regime, pure delta-rule recurrence cannot sustain measurable exact recall on genomic background at any distance, and an auxiliary decoupled memory — here the k-mer fast-weight table — is what restores it.* Whether a scaled-up pure KDA closes the gap without such a memory is untested here.

The $0.82 \rightarrow 0.62$ nat decay between 4 kbp and 64 kbp must be interference *inside* slot-level fast weights rather than address exhaustion, since address collision rates are distance-independent (§3.2) — Proposition 1's EMA horizon is the mechanism.

Degenerate motifs: the copy advantage survives point substitutions. Because the memory keys on *exact* 8-mers, the natural objection is that biological motifs are degenerate and a single SNP in a repeated segment could zero the mechanism out. We measured this directly: the probe re-run with m random point substitutions planted in the repeated 256 bp segment (`--needle-copy-mutations`; sequence identity 99.6% at m=1, 99.2% at m=2, 98.8% at m=3), for both memory arms at the full 131,072 context (Figure 10). For Mnemos-DNA the copy advan-

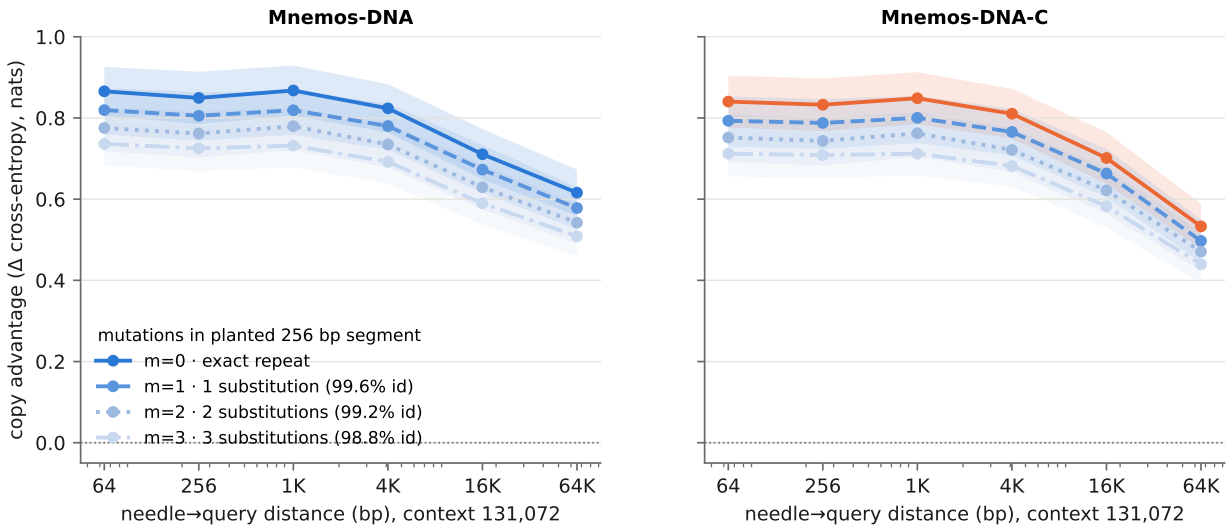


Figure 10: Recall under degenerate repeats (context 131,072, 128 trials/point, shaded 95% bootstrap bands). The repeat-copy probe of Figure 9 re-run with 0–3 random point substitutions planted in the repeated 256 bp segment, for Mnemos-DNA (left) and Mnemos-DNA-C (right). Each substitution costs ≈0.03–0.05 nats roughly uniformly across distances; the curves' shapes are unchanged, so the k-mer memory's recall is robust to SNP-level divergence between repeat copies — a substitution invalidates at most 8 of the segment's 249 8-mer addresses. All curves, m=0 through m=3, are measured on the main-suite checkpoints (step 34,177 for Mnemos-DNA, 33,900 for Mnemos-DNA-C).

tage falls from 0.87 to 0.74 nats at short range across m=0 → 3 and from 0.62 to 0.51 at 64 kbp; Mnemos-DNA-C behaves the same way (0.84→0.71 short range, 0.53→0.44 at 64 kbp). The cost is linear and small — ≈0.04 nats per substitution at short range, ≈0.03 at 64 kbp, with no threshold or collapse anywhere in the sweep — and the curve shapes are unchanged, so the decay with distance and the decay with divergence are independent. This is what the mechanism predicts: a substitution invalidates only the k=8 addresses overlapping it ($\leq$ 8 of 249 addressed positions per mutation), and at m=3 the remaining $\geq$90% of the segment's 8-mers still hit their slots. Exactness is required per *address*, not per *segment*.

4.7 Mechanism II: how the 65,536 slots are actually used

Table 6 answers the capacity question empirically rather than from the pigeonhole bound. Three things stand out. *The space is under-used relative to its own null*: 63.1% of slots touched where uniform sequence would touch 86.5%. *Collisions are the norm*: 68.5% of writes land on an occupied slot, above the 56.8% a uniform null predicts and well above the 50% pigeonhole floor — which §3.2 argues is survivable, since a slot retains up to 16 near-orthogonal key directions. And *the popular intuition about repeats is only half right*: poly-T and poly-A are indeed the top two slots by a wide margin, but the top 20 slots together carry under 1% of writes, so tandem repeats are not where the capacity goes. The gradient in write-gate strength with depth (0.23 at block 0 to 0.87 at block 6) is a learned division of labour — early layers accumulate over many occurrences of an 8-mer, late layers overwrite with the most recent one.

Per 131,072 bp held-out window (Mnemos-DNA, 8 windows)	Measured	Uniform-sequence null
Addressed writes	131,056	131,056
Distinct slots written (of $4^8 = 65{,}536$)	41,342 (63.1%)	56,661 (86.5%)
Colliding writes	68.5%	56.8%
Writes per written slot (mean / p99 / max)	3.17 / 15 / 234	2.31 / 6 / —
Effective slots $\exp(H)$ of the write distribution	29,779 (45.4%)	56,661
Share of writes in the top-20 slots	0.98%	—
of which low-complexity (period ≤ 4)	0.50%	—
Mean write gate g_w, block 0 $\rightarrow$ block 6	0.226 $\rightarrow$ 0.865	—

Table 6: Slot utilization on real hg38. Addresses are a deterministic function of the sequence, so the first five rows are identical across all 8 layers; only the write gate differs by depth. Genomic non-uniformity costs capacity: 36.9% of the address space is never touched in a 131 kbp window, against 13.5% under a uniform null, and the write distribution's effective support is 45.4% of the space. The most-written slot is poly-T (234 writes; poly-A is second at 138), but the repeat tail is *narrow* — the top 20 slots take under 1% of all writes — so the capacity loss is broad non-uniformity, not a few repeat expansions. Deeper layers write much harder than shallow ones, which under Proposition 1 means a shorter retention horizon per slot near the output.

4.8 Mechanism III: which layer carries the splice-site signal

This probe was designed to ask whether Mnemos-DNA-C's local-attention block exposes splice-site structure — the mechanism the unmatched-protocol numbers had appeared to demand (§4.2). The measurement stands on its own as a map of where splice-site information lives in the stack, and it anticipated the correction: it found nothing special at the window. We fit an independent frozen linear probe on each of the 8 blocks' pooled representations (one embedding pass, identical splits) on Nucleotide Transformer `splice_sites_all`:

Block	0	1	2	3	**4 (local attn)**	5	6	7
Probe accuracy	**0.689**	0.687	0.666	0.654	**0.653**	0.668	0.673	0.672

There is no separability jump at block 4 — it is the worst block of the eight, and linear separability is highest at block 0 and declines through the middle of the stack. Read alongside the matched fine-tuning result, the pieces now agree: the window exposes no emergent linearly decodable splice-site structure, and under a matched protocol the fine-tuned numbers show no repair for it to explain. The declining-with-depth profile is itself informative — splice-site information is most linearly accessible at the *embedding-adjacent* blocks and is progressively traded away through the stack, consistent with 131 K-scale pretraining allocating depth to long-range statistics at the expense of local grammar, which is the same damage the random-init control exposes (§4.2).

4.9 Deployment feasibility: memory and throughput

On the metric that matters for an edge deployment — the longest single sequence processable at all — our arms reach 1 Mbp at 5.6 GB, where the released checkpoints stop at 256 K (HyenaDNA-1M and -450K, OOM at 512 K) or 512 K (Caduceus-PH at 5.0 GB, Caduceus-PS at 7.6 GB; both OOM at 1 M) in the same harness on the same card; Figure 11 shows the batch-1 peak-VRAM curves, linear in length for every arm shown. Throughput is flat in length, as $O(N)$ requires (Figure 12).

Measurement protocol matters here as much as in fine-tuning: a batched embedding pipeline (8 sequences at once) yields apparent baseline ceilings of 16–32 K — an 8$\times$ overstatement of the gap.

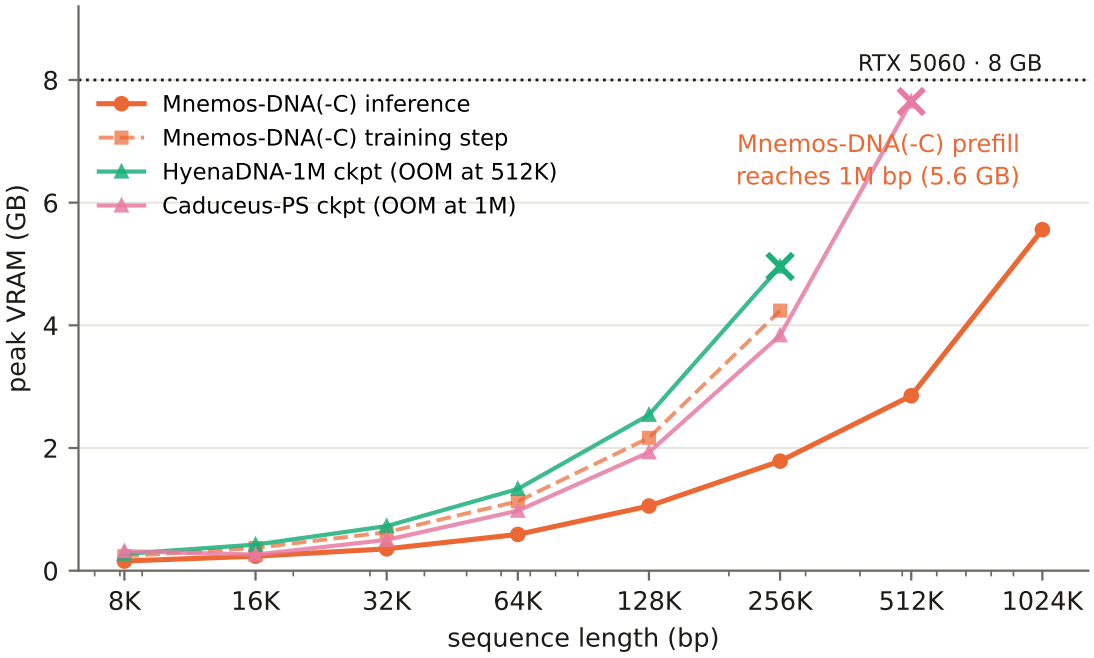


Figure 11: Peak VRAM versus sequence length (batch 1, bf16, single-sequence inference). Mnemos-DNA(-C) inference is linear and reaches 1 Mbp at 5.6 GB; a full training step fits through 256 K. The released baselines' own batch-1 curves are shown directly: HyenaDNA-1M reaches 256 K (5.0 GB) and goes out of memory at 512 K; Caduceus-PS reaches 512 K (7.6 GB) and goes out of memory at 1 M. **This is a deployment-feasibility comparison of released out-of-the-box checkpoints on consumer hardware**: the baselines (∼3× our parameter count) run as shipped, without a chunked path, while our 2.3M-parameter arms use a custom chunk-exact offload above 512 K. It measures what an 8 GB deployment of these artifacts can open — not an intrinsic sub-quadratic advantage of KDA over Hyena or Mamba, which this experiment cannot establish.

All ceilings reported in this paper come from single-sequence inference, the deployment-relevant measurement, under which our arms open windows 2–4× longer than the released baselines on this card.

Two honest qualifications remain. First, the comparison is between artifacts, not algorithms: those checkpoints are ∼3× our parameters (HyenaDNA-1M 6.5M, Caduceus-PH/PS 7.7M), are run as shipped without chunking, and were never intended for an 8 GB card, so the gap confounds model size, implementation and architecture. A parameter-matched, chunking-matched comparison is a different experiment and we have not run it. Second, the k-mer table makes our arms *less* memory-light than pure KDA (D01 runs a 1 Mbp forward in 5.5 GB natively and needs no offload anywhere in the sweep; Mnemos-DNA(-C) need the chunk-exact offload path above 512 K), which is the space cost of Table 1 showing up in practice.

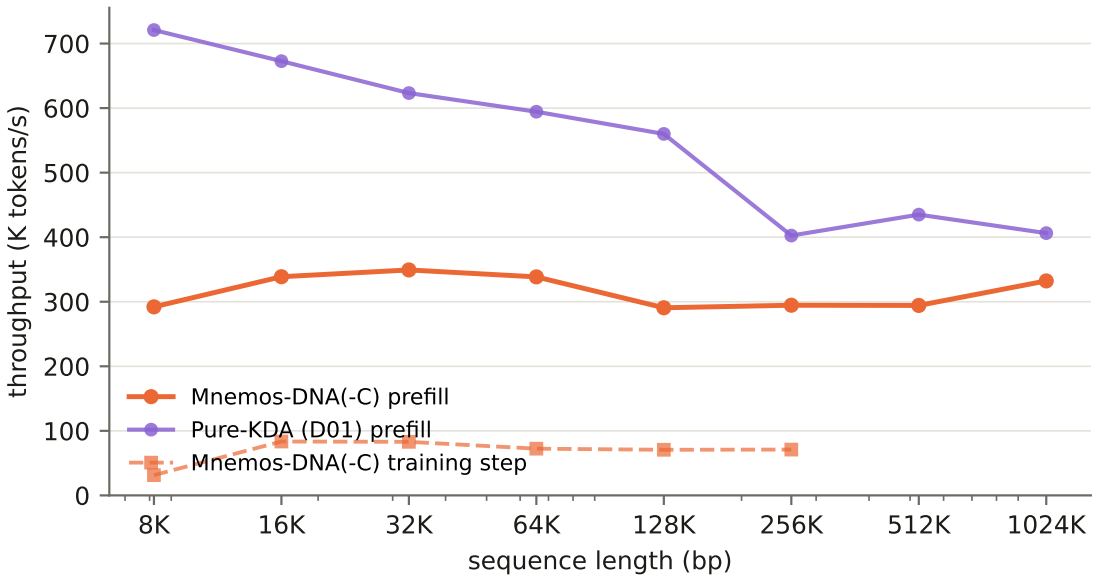


Figure 12: Throughput is flat in sequence length, as $O(N)$ requires: Mnemos-DNA(-C) prefill holds 250–350 K tok/s from 8 kbp to 1 Mbp, and a full training step holds ≈70 K tok/s. Against the matched Pure-KDA control (D01, 400–720 K tok/s) the k-mer table costs ≈2× prefill throughput on top of its VRAM cost (Table 1) — a real constant-factor price, though one that leaves the length-scaling flat (a prior comparison against the since-excluded D04 arm, whose inactive KV-cache path depressed its throughput, had understated this cost; Appendix B). Batch 1, bf16, one RTX 5060 (8 GB).

5 Discussion and Limitations

Protocol drift can manufacture architecture "results," and we measured it doing so. Two apparent headline findings — a +30.4-point splice repair and a −5.3-point GUE cost — arose during this study from comparing arms fine-tuned under different heads, epoch budgets and training-set caps, protocol drift accumulated over months; both vanished under the frozen matched protocol. We consider quantifying this the most transferable finding in the paper: fine-tuned downstream comparisons between architectures are fragile to exactly the kind of incremental protocol improvement (a better pooling head, a longer budget) that gets applied to the newest arm first. Everything downstream in this version runs under one frozen protocol, launched from a single queue script that records the full command line per row.

The composite is not decomposed — but it no longer needs to be. Mnemos-DNA-C's two changes were introduced together, so any Mnemos-DNA-C-specific effect belongs to the pair. Under the matched protocol the pair's downstream effect is ≤1 point on every suite mean, its recall effect is a modest reduction at long range, and its LM effect is 0.002 nats; the curriculum-only control is now of secondary interest, relevant mainly to whether short-context validation parity (Figure 3) can be had for free.

Seeds and intervals. Fine-tuned numbers now carry 2–3 seeds per cell (Table 3); typical per-task seed spread is 0.2–2 points, which is why we treat sub-2-point differences as unresolved even where bootstrap intervals are narrow. Pretraining remains a single run per arm at one data order — pretraining-seed variance is unmeasured, and the bootstraps still cover split-sampling noise only.

The large effects — the recall wall (0.6 nats), the random-init splice gap (∼9 points), the OOM ceilings — are far outside both noise terms.

Exact recall is per address, not per segment. The 4^8 address space keys on exact 8-mers, and biology's motifs are degenerate — the objection we expected to be sharpest. The mutation sweep (§4.6, Figure 10) now bounds it for both memory arms: up to three point substitutions in the repeated segment cost ≈0.04 nats each, linearly and without a threshold, and do not change the shape of the recall curve. What remains open is heavier divergence — indels, rearrangements, and substitution rates high enough to break most of the segment's 8-mers rather than a few percent of them.

Capacity ceiling at d_{model}=128. Nucleotide Transformer `promoter_all` is pinned at 0.84–0.85 for every arm and for the random-init control (GC content alone yields 0.81). At the edge-scale operating point this study targets, some tasks are simply out of reach, and we report them as capacity ceilings rather than as architecture comparisons.

Baseline comparisons are about artifacts. The HyenaDNA/Caduceus columns share our harness but not our model size (∼3× ours), pretraining budget, or inference path. Where they match or beat us the reading is unremarkable; where we match them at 2.3M parameters the reading is parameter efficiency; and their OOM ceilings describe 8 GB deployment of those released checkpoints, not the methods at their native scale — measured at batch 1, after we corrected an earlier probe-batch confound that had overstated the gap 8-fold (§4.9).

What we would test next. An intervention that actually closes the splice-site gap to random init — the window-plus-curriculum patch did not, so the candidates are shorter or mixed pretraining lengths from step 0, or a pretraining objective term that protects local grammar; the remainder of the mutation sweep (indels and rearrangements); a window-size sweep of the local layer; a bf16 fast-weight table to halve the 512 MiB inference state; and a variable k (or a learned address function) to attack the 36.9% of address space that hg38 never touches.

6 Conclusion

Gated delta-rule linear attention is a workable base operator for character-level genomic modeling at edge scale — but not alone. At this operating point's state capacity, pure KDA's finite-state recall bound is not a distant cliff on DNA; it is a wall at distance zero, and a windowed local-attention layer does not move it. A content-addressed k-mer fast-weight memory removes most of that wall — robustly to SNP-level divergence in the repeated material — for ∼3% extra parameters, a ≈2× prefill-throughput constant, and a 4096× larger recurrent state, while preserving $O(N)$ scaling and megabase-length exact chunked inference on an 8 GB GPU; all three costs are accounted for rather than omitted. Its address space is used unevenly: two thirds of slots touched in a 131 kbp window, 68.5% of writes colliding, and a delta rule inside each slot that turns those collisions into a gated regression rather than a clobber. The fixed ultra-long pretraining context that makes such models impressive at length makes them *worse* at sub-kilobase grammar than their own random initialization — a failure our window-plus-curriculum patch did *not* repair, and which stands as the study's sharpest open problem. And fine-tuned downstream comparisons proved protocol-fragile: apparent 30-point architecture effects arise from head, budget and data-cap drift alone and vanish under one matched fine-tuning protocol, which we report as a result in its own right: at this scale,

evaluation-protocol drift can produce larger architecture "effects" than architecture does. The trade-offs, not the wins, are the finding: recall, context length, and short-range grammar are three different budgets, and an architecture choice that spends one is spending it from somewhere.

A Pre-registered hypotheses and verdicts

The following hypotheses were stated in the study plan before the Mnemos-DNA(-C) comparison was run, so that outcomes could not be reframed after the fact. Each carries an explicit falsifier.

Hypothesis 1 (Deployment efficiency; high prior confidence)**.** *Under a fixed 8 GB VRAM ceiling, the $O(N)$ KDA arms will process strictly longer sequences than the released convolutional and SSM checkpoints, with peak memory growing linearly in sequence length.* Falsifier: *any baseline matching or exceeding the KDA arms' maximum feasible context on the same card.* Test: *§4.9.*

Hypothesis 2 (Recall bound)**.** *Pure KDA's exact-recall accuracy will degrade past a distance threshold set by its finite state capacity; the k-mer fast-weight memory, which factors exact 8-mer recall out of that state, will hold up at longer range; a windowed local layer adds exact recall only inside its ∼256–511 bp span.* Falsifier: *pure KDA matching the memory-augmented arms at long range, or the memory arms degrading identically to pure KDA.* Test: *§4.6.*

Hypothesis 3 (Language-model loss; no prior stated)**.** *Pretraining perplexity across arms was pre-registered as genuinely uncertain, with no directional expectation.* Test: *§4.1.*

Hypothesis 4 (Downstream capability; dissociation expected)**.** *Downstream fine-tuned accuracy was pre-registered as uncertain, with the explicit caution that perplexity may not predict it. A Mnemos-DNA-C-specific sub-hypothesis, formed after observing Mnemos-DNA's splice-site failure and before Mnemos-DNA-C was trained: a local-attention window plus a length curriculum will repair sub-kilobase position-sensitive tasks that fixed-131 K pretraining damaged.* Falsifier: *Mnemos-DNA-C failing to close the gap to the random-initialization control on splice sites.* Test: *§4.2.*

B The excluded D04 arm, and a bf16 cautionary note

Earlier drafts used an arm labelled D04 as the pure-KDA control. An audit found that D04 was in fact a KDA variant with an auxiliary evicted-KV exact cache whose 32 scalar cache gates, stored in bf16, sat at *exactly* their -4.0 initialization (sigmoid $= 0.018$) after the full 5.0B-token run: bf16's rounding granularity around small gradient updates froze them, so the cache path was inactive for the entire run while still costing throughput (its 80–98 K tok/s was the source of a prior, incorrect "the k-mer table is throughput-free" comparison). D04 was excluded and replaced by D01, a true pure-KDA arm trained under the fully matched protocol (NoPE, 131 K context, 4.5B tokens, same data order). The general lesson is the one already noted in §3.1: scalar gate parameters must be kept in fp32; every arm in this paper does so, and any arm trained before that fix should have its gate parameters checked against their initialization before its numbers are trusted.

References

[1] Eric Nguyen, Michael Poli, Marjan Faizi, Armin Thomas, Callum Birch-Sykes, Michael Wornow, Aman Patel, Clayton Rabideau, Stefano Massaroli, Yoshua Bengio, Stefano Ermon,

Hypothesis	Pre-registered prediction	Verdict	Evidence
H1 deployment efficiency	KDA arms exceed released baselines' max context on 8 GB	**Supported**, with a shrunk margin	1 Mbp vs. 256 K–512 K before OOM at batch 1 (Figs. 11, 6); an earlier 32 K/16 K figure was a probe-batch confound; confounds model size (∼3×)
H2 recall bound	Pure KDA degrades past a distance threshold; memory arms hold longer; a local window adds recall only inside its span	Supported, stronger than predicted	Matched pure KDA (D01) has *no* recall at any distance and the window-only arm (D02) is identical to it; memory arms hold to 64 kbp (Fig. 9), robustly to 1–3 substitutions (Fig. 10)
H3 LM loss	No direction stated	Parity	All arms within 0.01 nats; flat 8 K–1 M (Figs. 3, 4)
H4 downstream	Perplexity–capability dissociation; window+curriculum repairs sub-kb tasks	Dissociation supported; **repair sub-hypothesis falsified**	Under the matched protocol Mnemos-DNA-C does not close the gap to the random-init control on splice sites (0.733 vs. 0.832), so the stated falsifier fired. An earlier apparent +30.4-point repair was a fine-tuning-protocol artifact (§4.2); the GUE regression reported alongside it dissolved under the same audit

Table 7: Verdicts on the pre-registered hypotheses. "Supported" means the falsifier stated above did not occur — not that the proposed mechanism is confirmed. H4's repair sub-hypothesis is the study's one falsified prediction, and it was briefly "supported" by numbers that a protocol audit later invalidated — the sequence of events §5 argues is itself worth reporting.

Stephen A. Baccus, and Christopher Ré. Hyenadna: Long-range genomic sequence modeling at single nucleotide resolution. In *Advances in Neural Information Processing Systems 36 (NeurIPS)*, 2023. arXiv:2306.15794.

[2] Yair Schiff, Chia-Hsiang Kao, Aaron Gokaslan, Tri Dao, Albert Gu, and Volodymyr Kuleshov. Caduceus: Bi-directional equivariant long-range dna sequence modeling. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. arXiv:2403.03234.

[3] Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated delta networks: Improving mamba2 with delta rule. In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*, 2025. arXiv:2412.06464.

[4] Kimi Team. Kimi linear: An expressive, efficient attention architecture. *arXiv preprint arXiv:2510.26692*, 2025.

[5] Simran Arora, Sabri Eyuboglu, Aman Timalsina, Isys Johnson, Michael Poli, James Zou, Atri Rudra, and Christopher Ré. Zoology: Measuring and improving recall in efficient language models. *arXiv preprint arXiv:2312.04927*, 2023.

[6] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021. arXiv:2102.11174.

[7] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, et al. Nucleotide transformer: Building and evaluating robust foundation models for human genomics. *Nature Methods*, 22:287–297, 2025.

[8] Zhihan Zhou, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana Davuluri, and Han Liu. Dnabert-2: Efficient foundation model and benchmark for multi-species genome. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024. arXiv:2306.15006. GUE tasks obtained via the `leannmlindsey/GUE` HuggingFace community mirror.

[9] Michael Poli, Stefano Massaroli, Eric Nguyen, Daniel Y. Fu, Tri Dao, Stephen Baccus, Yoshua Bengio, Stefano Ermon, and Christopher Ré. Hyena hierarchy: Towards larger convolutional language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023. arXiv:2302.10866.

[10] Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *First Conference on Language Modeling (COLM)*, 2024. arXiv:2312.00752.

[11] Angelos Katharopoulos, Apoorv Vyas, Nikolaos Pappas, and François Fleuret. Transformers are RNNs: Fast autoregressive transformers with linear attention. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020. arXiv:2006.16236.

[12] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017.

[13] Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Wen Bo, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024. arXiv:2104.09864.

[14] Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. Yarn: Efficient context window extension of large language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*, 2024. arXiv:2309.00071. Describes the NTK-aware RoPE interpolation family.

[15] Valerie A. Schneider et al. Evaluation of GRCh38 and de novo haploid genome assemblies demonstrates the enduring quality of the reference assembly. *Genome Research*, 27(5):849–864, 2017.

[16] Laurens van der Maaten and Geoffrey Hinton. Visualizing data using t-SNE. *Journal of Machine Learning Research*, 9:2579–2605, 2008.

[17] Evan Trop, Yao Chen, McKinley Kang, Javier Mendoza-Revilla, Hugo Dalla-Torre, Bernardo P. de Almeida, and Thomas Pierrot. The genomics long-range benchmark: Advancing dna language models. *bioRxiv preprint*, 2024. InstaDeep Genomics Long-Range Benchmark (GLRB).

[18] Katarína Grešová, Vlastimil Martinek, David Čechák, Petr Šimeček, and Panagiotis Alexiou. Genomic benchmarks: A collection of datasets for genomic sequence classification. *BMC Genomic Data*, 24(1):25, 2023.